have been reported [14]. Also, no periodic smoothing has been applied to the distribution function; if periodic smoothing is effected as in [4] one can further decrease the number of "particles." This, together with a value of $\Delta t \sim 1$, can lead to an improvement of at least one order of magnitude on the performance actually presented.

Acknowledgment

The author wishes to thank Prof. G. Knorr for his assistance, and the authorities of the "Direction Générale de l'Enseignement Supérieur" for special permission to use the IBM 370/168 of the Ministry of Education of the Provincial Government of Québec (SIMEQ), as well as the SIMEQ authorities, who made it possible to carry out the calculations presented in this work.

References

[1] R. L. Morse and C. W. Nielson, *Phys. Rev. Lett.*, vol. 23, p. 1087, 1969.
[2] W. P. Gula and C. K. Chu, *Phys. Fluids*, vol. 16, p. 1135, 1973.
[3] H. R. Lewis and C. W. Nielson, *J. Comput. Phys.* vol. 17, p. 1, 1975.
[4] C-Z. Cheng and G. Knorr, *J. Comput. Phys.* vol. 22, p. 330, 1976.
[5] R. Gagné and M. Shoucri, *J. Comput. Phys.* vol. 24, p. 445, 1977.
[6] M. Shoucri and R. Gagné, *J. Comput. Phys.*, vol. 23, p. 242, 1977.
[7] T. M. O'Neil, *Phys. Fluids*, vol. 15, p. 103, 1974.
[8] G. J. Morales and T. M. O'Neil, *Phys. Rev. Lett.*, vol. 29, p. 914, 1972.
[9] V. L. Bailey and J. Denavit, *Phys. Fluids*, vol. 13, p. 451, 1970.
[10] G. Knorr, *Z. Naturforsch*, vol. A 18, p. 1304, 1963.
[11] J. M. Dawson and R. Shanny, *Phys. Fluids*, vol. 11, p. 1506, 1968.
[12] J. Canosa, *Phys. Fluids*, vol. 19, p. 1952, 1976.
[13] M. Shoucri and R. Gagné, *Phys. Fluids*, vol. 21, p. 1168, 1978.
[14] C-Z. Cheng, *J. Comput. Phys.*, vol. 23, p. 348, 1977.

Trajectories of Charged Particles in Radial Electric and Uniform Axial Magnetic Fields

GERALD W. ENGLERT

Abstract—**The trajectories of charged particles were determined over a wide range of parameters characterizing the motion in cylindrical low-pressure gaseous discharges and plasma-heating devices which have steady radial electric fields E perpendicular to uniform steady magnetic fields B. Three radial distributions of E were considered: $E \propto r$, constant E, and $E \propto r^{-1}$. These distributions are characteristic of the fields measured in a modified Penning discharge, in two NASA Lewis Burnout-type plasma-heating devices, and that estimated for the Ixion device, respectively. The plasmas of such $\vec{E} \times \vec{B}$ devices are often characterized by their high ratios of drift energy to mean particle energy, finite gyroradius effects, and sizeable electric field changes in the distance covered by a cyclotron radius. Such particle motions are not amenable to simple guiding center theory. From numerical calculations of the actual trajectories it was concluded that the differences between cyclotron frequency and qB/m, and between azimuthal drift and a guiding center approximation (including $\vec{E} \times \vec{B}$ and centrifugal force terms) are appreciable. The net cyclotron motion obtained by subtracting the actual drift from the trajectories, however, has a nearly circular contour over which the speed is quite constant.**

I. Introduction

TRAJECTORIES of charged particles in radial electric E and axial magnetic B fields have been of interest for many years in the study of gaseous discharges [1], [2] and more recently in the study of high-temperature plasma-heating devices [3]-[9]. Such devices often operate at low background pressure and relatively high magnetic fields where the ratio of cyclotron to large-angle collision frequencies is greater than 10^4. Under these conditions the free trajectories between inelastic and large-angle Coulomb collisions may determine the performance characteristics of a gaseous discharge, Hull [1]. The long-range interactions between charged particles are approximated by their influence on, or modification of, the electric field treated as a continuum.

It can be shown, by theory such as that of Morozov and Solov'ev [10], that the motion of charged particles in radial E and axial B fields can be approximated by a cyclotron motion superimposed on a curved drift about the field axis when 1) E/B is much less than the total of drift plus cyclotron velocity and 2) the spatial variations of the fields are very small over the span of the cyclotron motion. Under these conditions the drift can be approximated by E/B, and the cyclotron frequency by B times the charge-mass ratio of the particle. Arbitrarily small cyclotron radii are often assumed in guiding center theory, Grad [11].

Recent models of high-temperature plasmas, Englert *et al.* [12], [13], have been based on details of particle trajectories in order to interpret the influence of ion motion on neutral particle analyzer and optical monochromator measurements.

Manuscript received August 11, 1978.
The author is with the Lewis Research Center, National Aeronautics and Space Administration, Cleveland, OH 44135.

Results indicate that the thermalized component in the plane perpendicular to $\vec{B}$ is a randomized cyclotron velocity gyrating about a drift motion characterized by the electric and magnetic fields. The average particle energy, evaluated over an ensemble of particles in the plasma, is usually a modest fraction of the potential across the $\vec{E} \times \vec{B}$-type heating devices considered. As a result, mean thermal (cyclotron) and drift energy often have the same order of magnitude, and thus above condition 1) is violated.

That finite gyroradii have an effect on optical monochromator measurements of plasmas such as those in [5]-[9] has been suggested by Patch *et al.* [14] and studied by Englert *et al.* [13]. The electric fields in plasmas of interest often vary considerably (Hull [1], Boyer *et al.* [5], and Kambic [4]), over a distance equal to a cyclotron radius. Thus both of the above two conditions for use of a guiding center approximation are violated in these $\vec{E} \times \vec{B}$-type plasma-heating devices. The induced magnetic field due to low-density plasmas is usually negligible permitting B to be quite uniform.

In this investigation, azimuthal drift about the E-field axis of symmetry, actual cyclotron frequency, and the extent to which the net cyclotron motion (total motion in a trajectory minus drift) is constant in speed and approximates a circular orbit are determined from numerical calculation of the actual trajectories. Results are presented over wide ranges of ratio of drift to cyclotron frequencies, particle energy, and electric- and magnetic-field strengths.

All fields are assumed steady. Three E distributions were postulated: 1) E proportional to radius r from the field axis of symmetry, 2) uniform radial E, and 3) E inversely proportional to r. The distribution of case 1 closely fits the ion-beam probe measurements of Kambic [4] made on a modified Penning discharge operating in the low-pressure mode. It also corresponds to a charged cylinder with axis parallel to B, treated by Morozov and Solv'ev [10]. A uniform radial E (case 2) closely represents that obtained from Langmuir probe measurements on HIP and SUMMA (Burnout type) heating devices made by Englert, Patch, and Reinmann [13]. Case 3 is a distribution which was estimated for the Ixion by Boyer *et al.* [5]. It also corresponds to a line charge parallel to B (see e.g., Panofsky and Phillips [15]).

II. Analysis

A. Trajectory Equations

The trajectory equations [10] express time and azimuthal angle (Fig. 1) as functions of radius from the E-field axis of symmetry

$$t - t_i = \int_{r_i}^{r} \frac{dr}{\dot{r}} \tag{1}$$

and

$$\varphi - \varphi_i = \int_{r_i}^{r} \frac{\dot{\varphi}}{\dot{r}}\, dr \tag{2}$$

where

$$\dot{r} = \pm\left[2m^{-1}(\epsilon_{\perp,a} - q\Phi) - (r\dot{\varphi})^2\right]^{1/2} \tag{3}$$

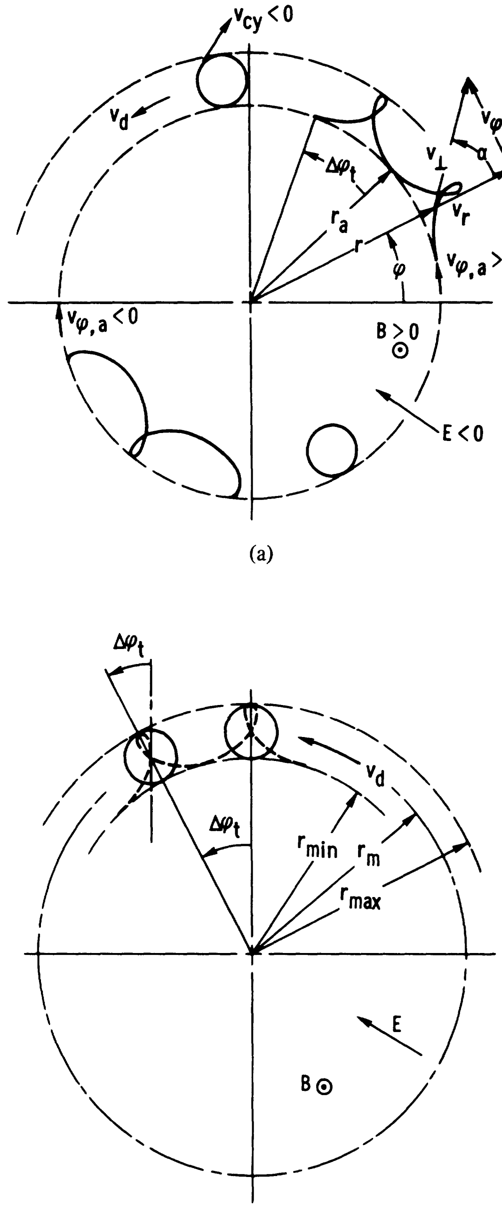

Fig. 1. Illustration of nomenclature. Examples are for ions with $E < 0$ and $B > 0$ causing counter-clockwise (positive) drift and clockwise cyclotron motion. (a) Trajectories starting at r_a for cases of $v_{\varphi,a} < 0$ and $v_{\varphi,a} > 0$. The net cyclotron motion is shown at the end of each trajectory. Dashed lines show radial limits. (b) Reorientation of net cyclotron orbit due to azimuthal drift, $\Delta\varphi_t$, about field axis of symmetry in one time period Δt during which a particle goes from the inner radial extrema to the outer one and back again.

and

$$\dot{\varphi} = p_\varphi/(mr^2) - \tfrac{1}{2}\,\omega_B. \tag{4}$$

The angular momentum, p_φ, is a constant of the motion. It is evaluated at a radial station, denoted by r_a, where both $\dot{r}$ and, arbitrarily, the electrostatic potential, Φ, are equal to zero. Integration of $\nabla\Phi = -E$ gives

$$\Phi = \begin{cases} -\tfrac{1}{2}\, E_a(r^2 - r_a^2)/r_a, & \text{for } E \propto r \\ -E_a(r - r_a), & \text{for constant } E \\ -E_a r_a \ln r/r_a, & \text{for } E \propto r^{-1}. \end{cases} \tag{5}$$

At r_a the kinetic energy in a plane perpendicular to $\vec{B}$ is denoted by $\epsilon_{\perp,a}$. The parameter $\omega_B \equiv qB/m$ is equal to the cyclotron frequency for a uniform B and either no or a uniform rectilinear E field [16]. Subscript i denotes an initial value. Usually in (1) and (2), $t_i = 0$, $\varphi_i = 0$, and $r_i = r_a$. SI units will be used throughout.

It is convenient to normalize the trajectory equations in terms of dimensionless variables R, Ω, and $\mathcal{E}$ which are defined as

$$R \equiv r/r_a \tag{6}$$

$$\mathcal{E} \equiv (\epsilon_{\perp,a} - q\Phi)/(qE_a r_a) \tag{7}$$

and

$$\Omega \equiv E_a/(Br_a\omega_B). \tag{8}$$

The parameter $\mathcal{E}$ has the form of a kinetic energy divided by a potential energy. The value of $\mathcal{E}$ at $r = r_a$ is denoted by $\mathcal{E}_a$. The parameters $\mathcal{E}_a$ and Ω are constants for any specific trajectory. For low E_a/B and high r_a, Ω is approximately the frequency of rotation about the E-field axis of symmetry divided by the cyclotron frequency, as will be shown later.

Use of these dimensionless ratios and substitution of (3) to (5) into (1) and (2) yields

$$t = 2\omega_B^{-1}\int \frac{RdR}{\mathcal{F}(R)} \tag{9}$$

and

$$\varphi = C\int \frac{R^{-1}dR}{\mathcal{F}(R)} - \frac{1}{2}\omega_B t \tag{10}$$

where

$$\mathcal{F}(R) = \begin{cases} \{-(1-4\Omega)R^4 + 2[4\Omega(\mathcal{E}_a - \frac{1}{2}) + C]R^2 - C^2\}^{1/2}, & \text{for } E \propto R \\ \{-R^4 + 8\Omega R^3 + [2C + 8\Omega(\mathcal{E}_a - 1)]R^2 - C^2\}^{1/2}, & \text{for constant } E \\ \{-R^4 + [8\Omega\mathcal{E}_a + 2C + 8\Omega \ln R]R^2 - C^2\}^{1/2}, & \text{for } E \propto R^{-1} \end{cases} \tag{11}$$

and $C = 1 \pm (8\Omega\mathcal{E}_a)^{1/2}$. The two values of C correspond to the two solutions of

$$v_{\varphi,a} = \pm(2\epsilon_{\perp,a}/m)^{1/2}. \tag{12}$$

There are thus two trajectories for each combination of Ω and $\mathcal{E}_a$. They are labeled as $v_{\varphi,a} < 0$ and $v_{\varphi,a} > 0$ on Fig. 1.

For each trajectory the equation $\mathcal{F}(R) = 0$ has two positive real roots[1] (where $\dot{r} = 0$) for all three radial distributions of E.

[1]That $R = 1$ is a root can be verified by factoring (11). For the case of $E \propto R$ (11) is quadratic in $X = R^2$ and the second root is found immediately in the quotient of $\mathcal{F}(R)$ divided by $X - 1$. A polynomial root finding method [17] and the Newton-Raphson method [18] were used to find the second real roots for the cases of constant E and $E \propto R^{-1}$, respectively.

B. Drift and Characteristic Frequencies

Azimuthal drift for this periodic motion can be defined as $\Delta\varphi_t/\Delta t$ where $\Delta\varphi_t$ (Fig. 1(a)) is the change of azimuthal angle, and Δt the time spent as a particle goes from one radial extremum to the other and back again. The drift ratio

$$D \equiv -(\Delta\varphi_t B r_m)/(E_a\Delta t) \tag{13}$$

can be viewed as the actual guiding center velocity $r_m\Delta\varphi_t/\Delta t$ (which accounts for the cylindrical geometry and the dependence of E on r) divided by the approximation $-E_a/B$. Here, the mean radius is $R_m = 1/2(R_{\max} \pm R_{\min})$. The positive sign applies to all conditions except when both $v_{\varphi,a} < 0$ and $\Omega < (8\mathcal{E}_a)^{-1} < 0$, which occur when the trajectory (as will be shown later) encircles the E axis of symmetry.

The ratio F of the azimuthal drift frequency $\Delta\varphi_t/(2\pi\Delta t)$ about the E-field axis, to the frequency $(\Delta t)^{-1}$ of completing cyclotron orbits between $R_{\min}$ and $R_{\max}$ is

$$F = \pm\Delta\varphi_t/2\pi. \tag{14}$$

The sign of F is positive if drift and cyclotron rotation have the same direction. Thus the sign of F is the same as that of Ω.

The ratio f of cyclotron frequency to ω_B is

$$f = 2\pi/(\omega_B\Delta t). \tag{15}$$

The change in orientation of these inner cycles (Fig. 1(b)) during Δt can be accounted for, if desired, by multiplying f by $(1 + F)$.

For the case of $E \propto R$ the integrations of (9) and (10) between the radial extrema $R_{\min}$ and $R_{\max}$ can be performed analytically.[2] There results

$$\Delta t = 2\pi/[\omega_B(1 - 4\Omega)^{1/2}] \tag{16}$$

$$\Delta\varphi_t = \pi[1 - (1 - 4\Omega)^{-1/2}] \tag{17}$$

and thus by (13)

$$D = \tfrac{1}{2} R_m\Omega^{-1}[1 - (1 - 4\Omega)^{1/2}]. \tag{18}$$

These results show that F and f are independent of $\mathcal{E}_a$ and $v_{\varphi,a}$ for the case of $E \propto r$. The ratio D, however, depends on these quantities through R_m.

Numerical integration of (9) and (10) was used for the cases of constant E and $E \propto R^{-1}$, and even for the case of $E \propto R$ when integration is not restricted to complete half cycles between $R_{\min}$ and $R_{\max}$. Special care is now required in the vicinities of the R extrema when the integrand becomes singular.[3]

[2]Let $X = R^2$ and R^{-2} in the integrals for Δt and $\Delta\varphi_t$, respectively. Then apply, for example, equation (161) of Peirce [19].

[3]For the case of $E \propto R^{-1}$ the logarithm in (11) is expanded locally in a power series. Such improper integrals for all three E-field distributions can then be placed in the form

$$\int_0^x x^{-1/2}g(x)\,dx$$

about the lower and upper extrema. This is accomplished by factoring $\mathcal{F}(R)$ and letting x equal $1/2(R - R_{\min})/(R_{\max} - R_{\min})$ and $1/2(R_{\max} - R)/(R_{\max} - R_{\min})$ and modifying $g(x)$ appropriately. The integrals can then be evaluated by use of methods described in [20].

C. Trajectory Contours and Net Cyclotron Motion

Integration of (9) and (10) with one of the limits of integration held fixed and the other varied yields $\varphi(R)$ and $t(R)$, and thus $\varphi(t)$. The net cyclotron trajectory can be obtained by plotting $\varphi - (\Delta\varphi_t/\Delta t)t$ versus R.

The magnitude of the net cyclotron velocity v_{cy}, over an orbit can be expressed as

$$v_{cy}^2 = (\vec{v}_\perp - \vec{v}_d) \cdot (\vec{v}_\perp - \vec{v}_d)$$

where $\vec{v}_\perp$ is the particle velocity in the $\perp$ plane and $v_d \equiv r_m \Delta\varphi_t/\Delta t$. Use of conservation of energy yields

$$v_{cy}/(\omega_B r_a) = [(R - CR^{-1})D\Omega + (D\Omega)^2 + 2\Omega\mathcal{E}]^{1/2} \tag{19}$$

where from (5)

$$\mathcal{E} = \begin{cases} \mathcal{E}_a + \frac{1}{2}(R^2 - 1), & \text{for } E \propto r \\ \mathcal{E}_a + R - 1, & \text{for constant } E \\ \mathcal{E}_a + \ln R, & \text{for } E \propto R^{-1}. \end{cases} \tag{20}$$

D. Identification of Trajectories by Parameters Other Than Ω, $\mathcal{E}_a$, and $v_{\varphi,a}$

In some problems of interest, initial conditions are selected by specifying the energy $\epsilon_\perp$, direction $\alpha + \varphi$, and location (r, φ) of a charged particle's motion in a certain electromagnetic field. To use results herein at the fullest advantage it is then necessary to find Ω, $\mathcal{E}_a$, and r_a as functions of $\epsilon_\perp$, α, r, E, B, and q/m.

The direction α (Fig. 1(a)) can be expressed as

$$\cot(\alpha) = \dot{r}/v_\varphi = \pm\{[8\mathcal{E}\Omega R^2/(C - R^2)^2] - 1\}^{1/2}.$$

Combining this with (20) eliminates $\mathcal{E}_a$ from C and in principle allows for solution of r_a in terms of the desired variables. For example let $\xi \equiv \epsilon_\perp/(qEr)$ and $\psi \equiv E/(B\omega_B r)$. Then for the case of $E = E_a r/r_a$,

$$r_a = [\mathfrak{B} \pm (\mathfrak{B}^2 - \mathfrak{A})^{1/2}]^{1/2}$$

where

$$\mathfrak{A} = [1 \pm (8\xi\psi)^{1/2} \sin\alpha]^2 r^4/(1 - 4\psi)$$

$$\mathfrak{B} = r^2\{\xi - \tfrac{1}{2} + [1 \pm (8\xi\psi)^{1/2} \sin\alpha]/(4\psi)\}/[(4\psi)^{-1} - 1].$$

The cases for constant E and $E \propto r^{-1}$ are considerably more difficult but can be solved numerically for r_a by methods such as that of [18]. Once r_a is known, Ω and $\mathcal{E}$ follow immediately and $\mathcal{E}_a$ can be found from (20).

E. Ranges of Ω and $\mathcal{E}_a$ of Interest

The formation of strong electric fields in suitably large volumes of plasma is usually made possible by the use of strong magnetic fields. The average particle energy is usually a modest fraction of the potential across the plasma-heating device. The value of $|\mathcal{E}|$ for ions in [3] to [9] ranges from about 0 to 0.4. The value of $|\Omega|$, in turn, ranges mostly from 0-0.1 and seldom gets as large as 0.2. Values of $|\Omega|$ of interest for electrons cover a much lesser range due to their smaller mass unless they are concentrated in regions of much smaller r_a. The range of $\mathcal{E}_a$ for ions and inward-directed E fields is from about -0.7 to 0 for $v_{\varphi,a} > 0$ and about -0.4–0 for $v_{\varphi,a} < 0$.[4]

III. Results and Discussion

A. Radial Extrema of Trajectories

The two real roots R_{ex} of (11) which are the radial extrema, R_{min} and R_{max}, of the trajectories are plotted versus Ω for fixed $\mathcal{E}_a$ on Fig. 2. For ions and inward $\vec{E}$, or electrons and outward $\vec{E}$, both Ω and $\mathcal{E}_a$ are negative. For ions and outward $\vec{E}$, or electrons and inward $\vec{E}$, both Ω and $\mathcal{E}_a$ are positive. Since B enters into only the Ω parameter and then as a square, its sign does not influence the results. The solid and dashed curves correspond to the positive and negative solution of $v_{\varphi,a} = \pm\sqrt{2m\epsilon_{\perp,a}}$.

As $R_{ex} = 1$ is always a root, the radial span of the total trajectory as well as the net cyclotron motion is normally the difference between the $R_{ex} \neq 1$ root and unity. The exception is when the trajectory encompasses the axis of field symmetry and its radial span is then the sum of the $R_{ex} \neq 1$ root and unity. This occurs when $v_{\varphi,a} < 0$ and $\Omega < (8\mathcal{E}_a)^{-1} < 0$ simultaneously. The trajectory intersects the axis of field symmetry when $v_{\varphi,a} < 0$ and $\Omega = (8\mathcal{E}_a)^{-1}$. At these points results are questionable for two of the E-field distributions since the uniform E assumption breaks down physically and the $E \propto r^{-1}$ assumption leads to a singularity here.

For $v_{\varphi,a}$ and Ω of like sign the radial span of the trajectory increases with increase of $|\Omega|$. For $v_{\varphi,a}$ and Ω of opposite sign a maximum radial span is reached as $|\Omega|$ is increased from its zero value. As $|\Omega|$ is increased further the radial span decreases until $R_{min} = R_{max} = 1$. Here the trajectory reduces to a circle about the field axis of symmetry. To find this location, appeal is made to the radial component of the momentum equation,

$$m\ddot{r} = mr\dot{\varphi}^2 + q(E + r\dot{\varphi}B). \tag{21}$$

When $\ddot{r} = 0$ and $r = r_a$, the azimuthal velocity is

$$r_a\dot{\varphi} = -\tfrac{1}{2}\omega_B r_a[1 - \sqrt{1 - 4E/(B\omega_B r_a)}] \tag{22}$$

for the case of interest on Fig. 2. This can be expressed as

$$\mathcal{E}_a = (1 - \sqrt{1 - 4\Omega})^2/8\Omega \tag{23}$$

and is independent of the E-field distribution. The lower the value of $|\mathcal{E}_a|$ the lower the value of $|\Omega|$ for an intersection of the constant $\mathcal{E}_a$ lines with the $R = 1$ line. The maximum value of Ω for such an intercept is $\frac{1}{4}$. For the E proportional to r-field distribution the curves of constant $\mathcal{E}_a > 0$ are asymptotic to the $\Omega = \frac{1}{4}$ line.

B. Trajectory Contours

Trajectories corresponding to various Ω and $\mathcal{E}_a$ and the three E distributions of Fig. 2 are shown in Fig. 3. Two complete cycles, each consisting of paths from one radial extremum to the other and back again, are given for each trajectory on this figure. This is to better illustrate the periodic nature

[4] In Fig. 1(a), Ω and $\mathcal{E}_m$ (i.e., $\mathcal{E}$ at R_m) are fixed at values of -0.11 and -0.14, respectively, for both trajectories illustrated. Here $\mathcal{E}_a$ is equal to -0.10 and -0.27 for $v_{\varphi,a} < 0$ and $v_{\varphi,a} > 0$, respectively.

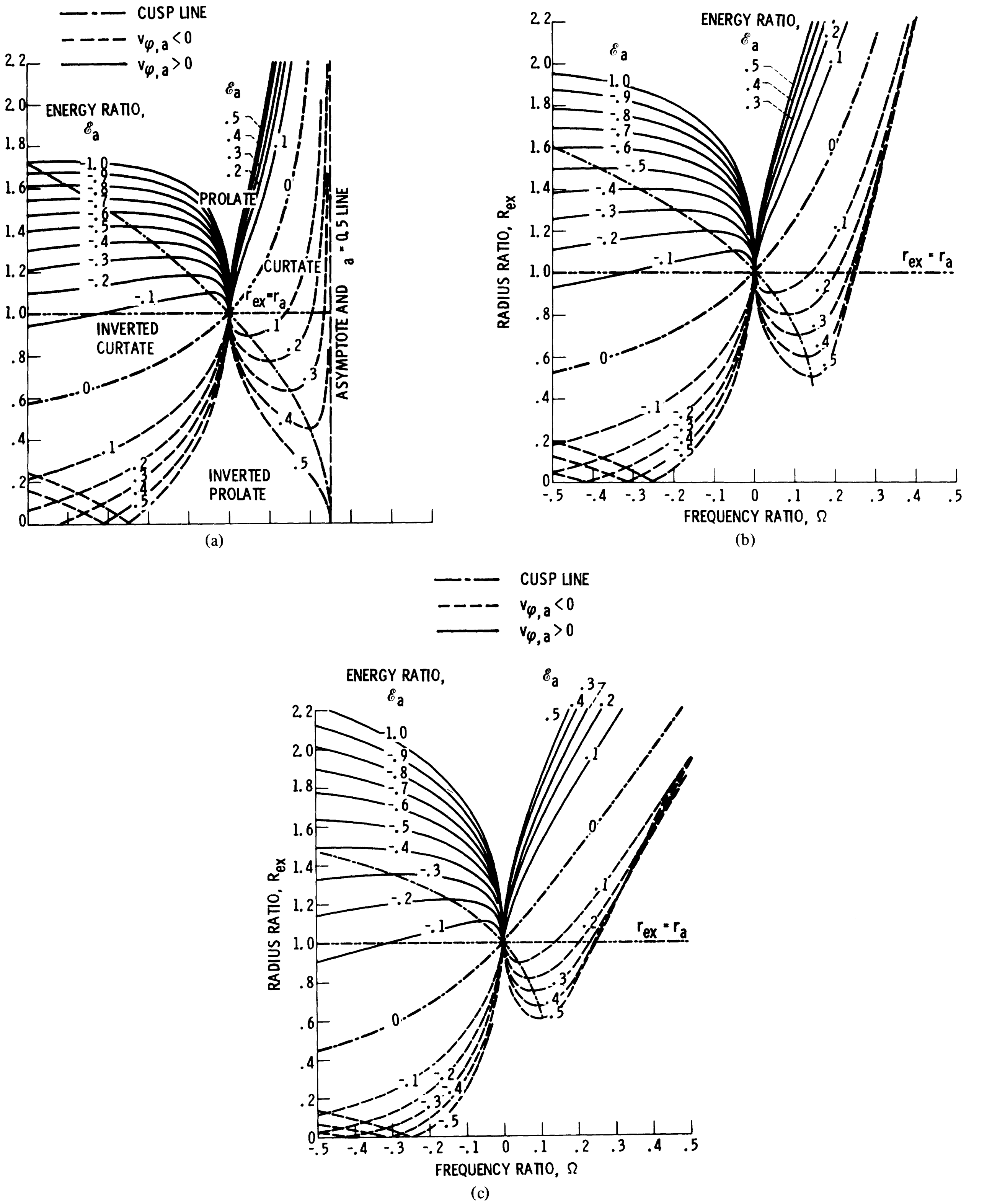

Fig. 2. Radial extrema, $R_{ex} = R_{min}$ and R_{max}, of the trajectories. (a) E proportional to R. (b) Constant E. (c) E inversely proportional to R.

of the trajectories as they curve about the axis of field symmetry. The (arbitrary) starting point for each inner (cyclotron) cycle is indicated by the data symbol. Only one cycle of the net cyclotron motion is given. The φ values of the starting points were kept quite arbitrary for ease of programming the plotting routine.

By borrowing terminology descriptive of a circle rolling on a fixed circle, the trajectories can be classified as ones which resemble epicycloids in general appearance. For example, the trajectory of Fig. 3(b) with its sharp cusps pointing inward resembles an epicycloid, whereas the trajectories of parts (a) and (c) resemble prolate and curtate epicycloids, respectively.

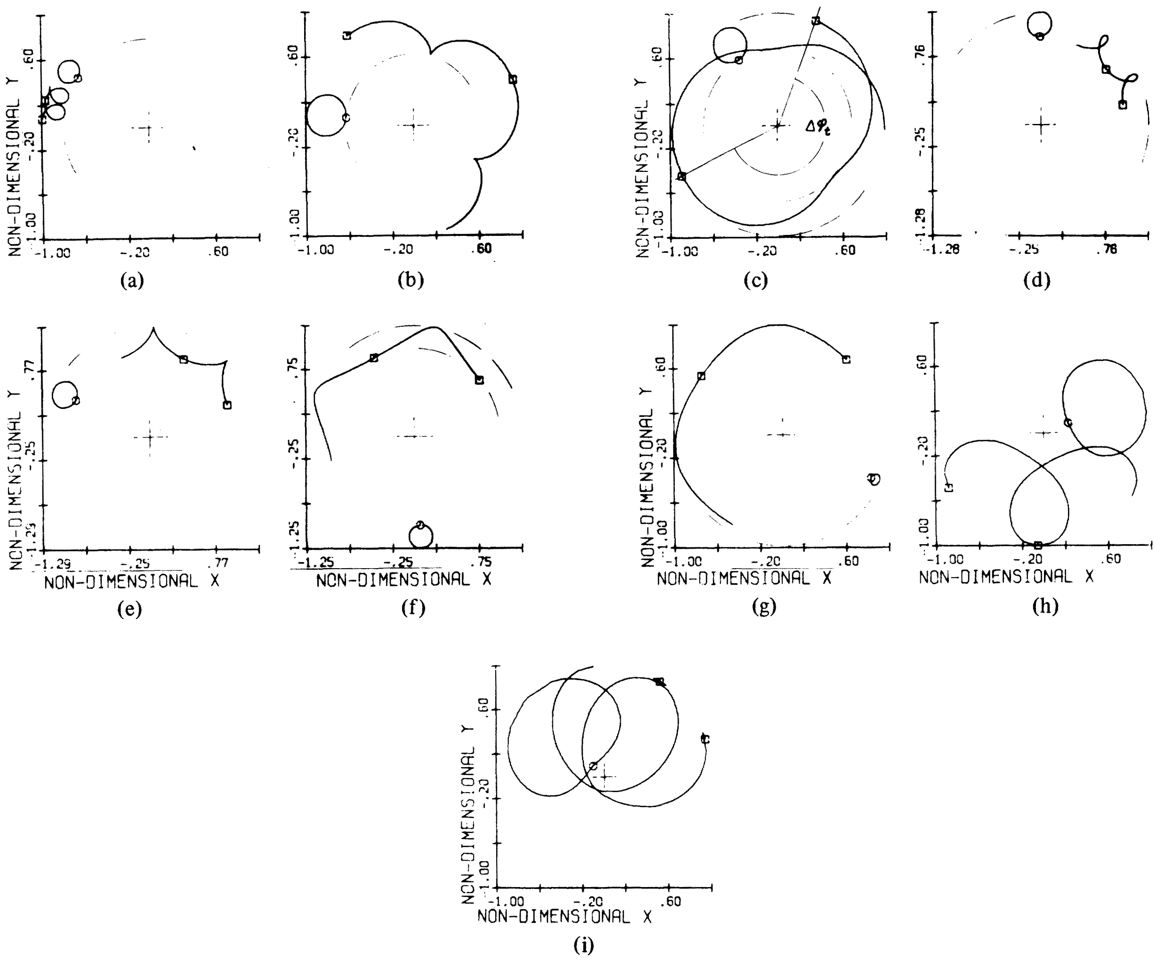

Fig. 3. Representative trajectories. Parts (a), (b), and (c) are with $E \propto R$; (d), (e), and (f) are with constant E; and (g), (h), and (i) are with $E \propto R^{-1}$. Dashed lines added to show radial limits.

(a) $v_{\varphi,a} < 0$, $\mathcal{E}_a = 0.3$, $\Omega = 0.025$, $D = 0.92$, $|\Delta v_{cy}/v_{cy}| = 0.026$
(b) $v_{\varphi,a} < 0$, $\mathcal{E}_a = 0.3$, $\Omega = 0.150$, $D = 1.00$, $|\Delta v_{cy}/v_{cy}| = 0.226$
(c) $v_{\varphi,a} < 0$, $\mathcal{E}_a = 0.3$, $\Omega = 0.200$, $D = 1.16$, $|\Delta v_{cy}/v_{cy}| = 0.382$
(d) $v_{\varphi,a} > 0$, $\mathcal{E}_a = -0.3$, $\Omega = -0.1$, $D = 0.92$, $|\Delta v_{cy}/v_{cy}| = 0.040$
(e) $v_{\varphi,a} > 0$, $\mathcal{E}_a = -0.3$, $\Omega = -0.2$, $D = 0.86$, $|\Delta v_{cy}/v_{cy}| = 0.071$
(f) $v_{\varphi,a} > 0$, $\mathcal{E}_a = -0.3$, $\Omega = -0.5$, $D = 0.75$, $|\Delta v_{cy}/v_{cy}| = 0.134$
(g) $v_{\varphi,a} > 0$, $\mathcal{E}_a = -0.1$, $\Omega = -0.5$, $D = 0.75$, $|\Delta v_{cy}/v_{cy}| = 0.041$
(h) $v_{\varphi,a} < 0$, $\mathcal{E}_a = -0.5$, $\Omega = -0.1$, $D = 1.25$, $|\Delta v_{cy}/v_{cy}| = 0.103$
(i) $v_{\varphi,a} < 0$, $\mathcal{E}_a = -0.5$, $\Omega = -0.5$, $D = 0.18$, $|\Delta v_{cy}/v_{cy}| = 0.521$

For an illustration of true epicycloids see, for example, [21]. Although the net cyclotron motion is quite circular, the angular-drift velocity is too low in (a) and too high in (c) for use of the rolling-cylinder terminology in the true sense. The trajectories in (d)-(f) resemble cylinders rolling on the inside of larger circles and will be, therefore, dubbed "inverted" epicycloids.

The $\mathcal{E}_a = 0$ curves on Fig. 2(a)-(c) can be identified as the locus of points where the trajectories have sharp cusps. These cusps point outward for $\Omega < 0$ and inward for $\Omega > 0$. Curves connecting the maximum R_{ex} values of the constant $\mathcal{E}_a$ curves for $\Omega < 0$ and the minimum values for $\Omega > 0$ are also loci of cusps and are shown on Fig. 2(a)-(c) by heavy dot-dash lines. The intersection of these two curves separate the R_{ex}, Ω space into four zones in which the trajectories roughly have the nature of "regular" prolate and curtate epicycloids and "inverted" prolate and curtate epicycloids.

Part (f) and (g) of Fig. 3 illustrate the continuing curtate nature of the trajectories for $v_{\varphi,a} > 0$ as Ω is held constant at -0.5 and $\mathcal{E}_a$ is varied sufficiently to cross the $R_{ex} = 1$ line in R_{3x}, Ω space. The trajectories are circles where $\mathcal{E}_a = -0.1338$, which is in agreement with (23).

Parts (h) and (i) of Fig. 3 illustrate the continuing prolate nature of the trajectories for $v_{\varphi,a} < 0$ as $\mathcal{E}_a$ is held constant at -0.5 and Ω is decreased from greater than to less than $(8\mathcal{E}_a)^{-1}$. Part (i) shows that the trajectories encompass the axis of field symmetry for $\Omega < (8\mathcal{E}_a)^{-1} < 0$. The span of the net cyclotron motion here is the sum $R_{max} + R_{min}$. The arithmetic mean radii of the whole trajectory (which is used in D) is equal to one half of this sum.

The net cyclotron component of a trajectory closely resembles a circle. The spread of the magnitude of cyclotron velocity over an orbit $|\Delta v_{cy}/v_{cy,max}|$ decreases as $|\Omega|$ and $|\mathcal{E}_a|$ decrease. This deviation is listed on Fig. 3. From numerous plots, not shown, it is apparent that the separation of the motion into a drift and a *circular* cyclotron velocity of constant magnitude is most justifiable for the E proportional to r case, and almost as good for the uniform E case. It is noticeably less accurate for the E inversely proportional to r case but still very reasonable. The largest deviation is expected for this latter case since E can become especially large in the vicinity of $R = 0$.

The distance between data points is always connected by straight lines on these computer plots. This causes apparent flat spots on the curves at times. This is especially noticeable in the net cyclotron orbits in Fig. 3(h) and (i) where the spacing between points is large in certain areas. The spacing between calculated points however does not affect the accuracy of the calculation at the points.

C. Frequency Ratios F and f, and Drift Ratio D

The ratios F and f defined by (14) and (15) are given on Figs. 4 and 5, respectively. Both of these ratios are quite dependent on $\mathcal{E}_a$ and $v_{\varphi,a}$ except for the case of $E \propto r$. This dependence is larger for $\Omega > 0$ than for $\Omega < 0$.

At low values of $|\Omega|$ and $|\mathcal{E}_a|$, Ω is a close approximation to F. In the range of most interest in [5]-[9], f lies within the range of 1.0 to 1.35, 1.4, and 1.55 for $E \propto r$, constant E, and $E \propto r^{-1}$, respectively.

The ratio D, defined by (13) is plotted in Fig. 6(a)-(c) for the three radial E-field distributions. In the Ω, $\mathcal{E}_a$ region of most interest for ions in [5]-[9], D lies within 0.49 and 1.23.

Since the location where $\Phi = \Phi_a$ is arbitrary, r_a can be selected in the range of the application of interest. If, for example, r_a is increased while holding $\epsilon_{\perp,a}$, E, and B fixed, the absolute values of Ω and $\mathcal{E}_a$ on Figs. 2-6 are decreased. As Ω and $\mathcal{E}_a$ approach 0, F also approaches 0, whereas R_{ex}, f, and D approach 1. Thus the trajectories reduce to the case of a rectilinear $\vec{E}$ field. If on the other hand, Ω, $\mathcal{E}_a$, and E are held constant while increasing r_a (by increasing $\epsilon_{\perp,a}$ and decreasing B) the span of the cyclotron orbit increases in direct proportion to r_a. Here, of course, the ratios f, F, and D remain constant. The true cyclotron frequency and the frequency of rotation about the field axis of symmetry now decrease while drift velocity increases.

Drift is sometimes approximated [5] by setting $\ddot{r} = 0$ and $r\dot{\varphi} = v_d$ in (21) and letting $r = r_m = r_a$ as in a guiding center approximation. This is equivalent to setting v_d equal to $r_a\dot{\varphi}_a$ in (22). For the E proportional to r case this is equivalent to merely setting $r_m = r_a$ in (18). This approximation offers an improvement over the simple E/B approximation by considering centrifugal force as well. In dimensionless variables it can be written as

$$D_{\text{approx}} = (1 - \sqrt{1 - 4\Omega})/(2\Omega) \tag{24}$$

and its numerical evaluation is shown by the curves comprised of short dashes on Fig. 6(a)-(c). Equation (24) usually approximates the drift curves better than the E/B approximation, especially for the case of constant E. Its deficiency is most apparent where the span of the cyclotron motion ($r_{max} - r_{min}$) is large compared to the radius of curvature of the drift motion.

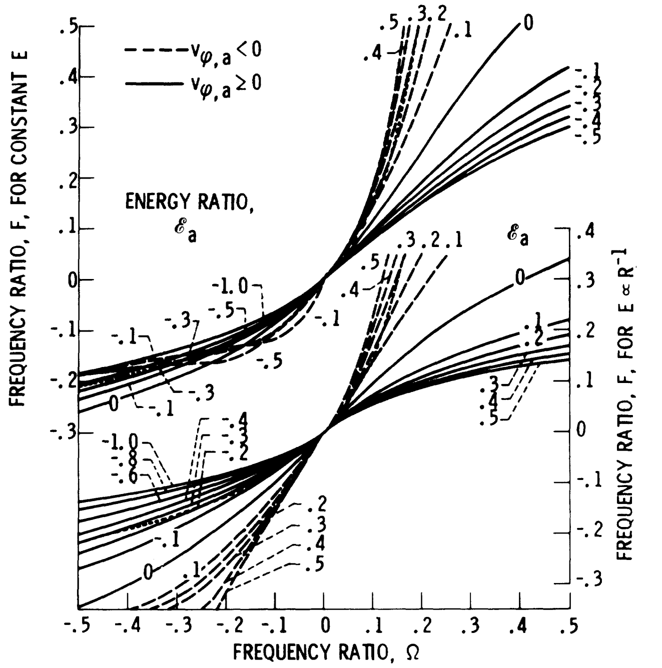

Fig. 4. Ratio, F, of azimuthal drift about E-field axis of symmetry to angular cyclotron motion about guiding center. Top set of curves are for constant E; bottom set are for $E \propto R^{-1}$. Case of $E \propto R$ is given by · · · · .

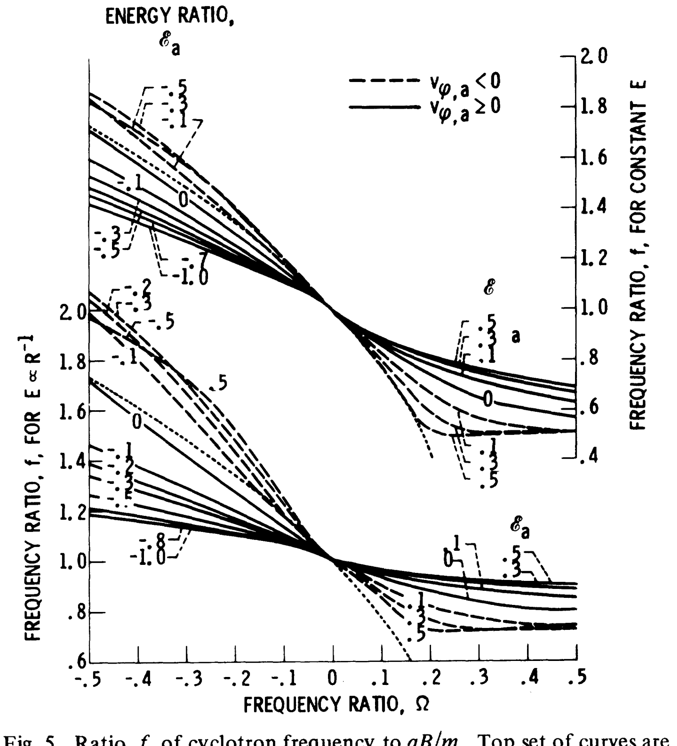

Fig. 5. Ratio, f, of cyclotron frequency to qB/m. Top set of curves are for constant E; bottom set are for $E \propto R^{-1}$. Case of $E \propto R$ is given by · · · · .

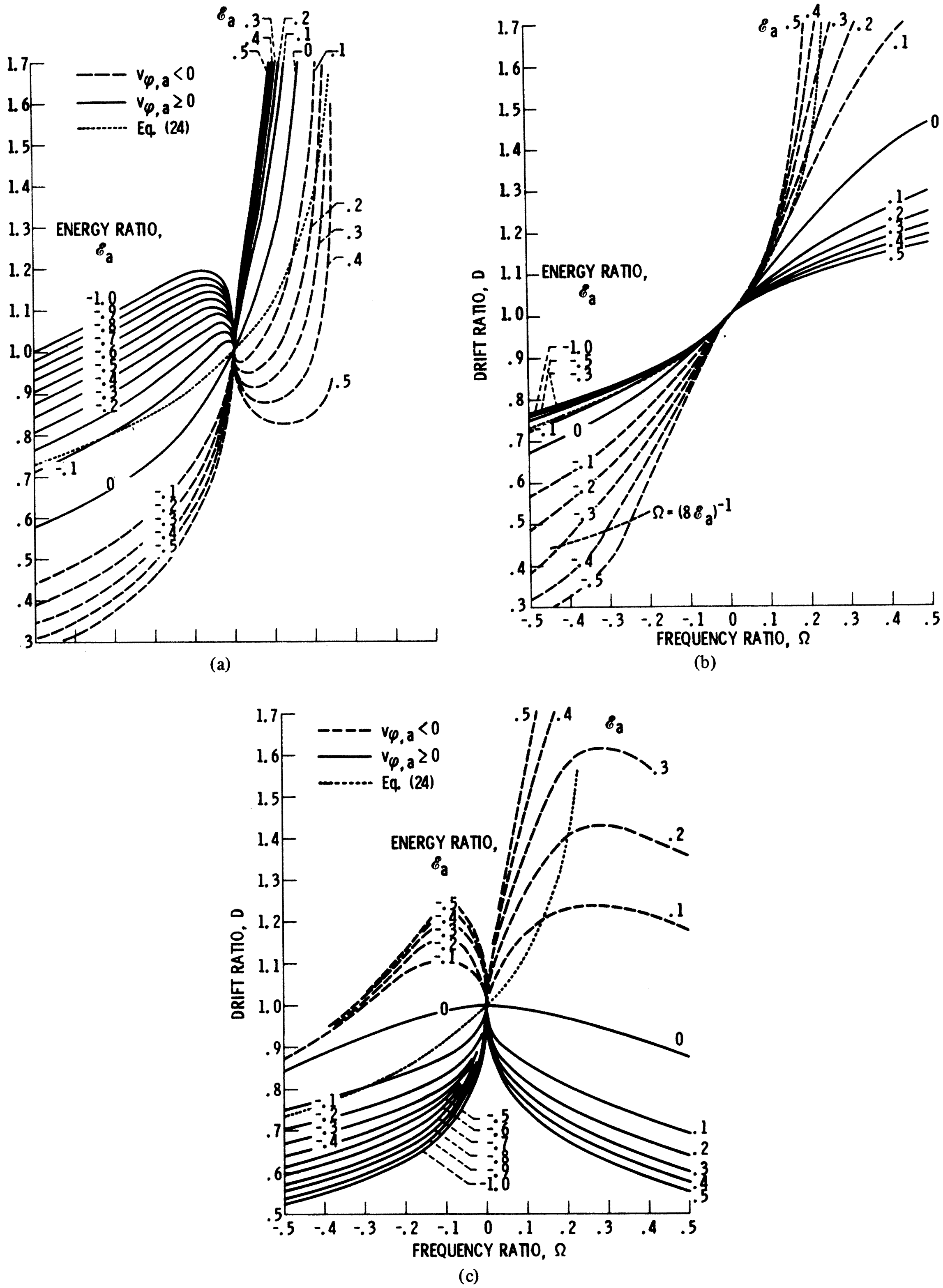

Fig. 6. Dependence of ratio, D, of azimuthal drift velocity to $-E/B$ on energy and frequency ratios. (a) E proportional to R. (b) Constant E. (c) E inversely proportional to R.

IV. Conclusions

From this study of the motion of charged particles in steady uniform axial magnetic fields B and three radial distributions of steady radial electric fields E, the following conclusions were reached.

1) Identification of the trajectories of the particles by frequency ratio, $\Omega \equiv (E_a/Br_a)/(qB/m)$, and energy ratio $\mathcal{E}_a \equiv \epsilon_{\perp,a}/(qE_ar_a)$, parameters and the sign of azimuthal velocity $v_{\varphi,a}$ at r_a generalize results as well as aid in the interpretation of results. Here $\epsilon_\perp$, r, m, and q are the energy, radial location,

mass and charge of a particle. The arbitrary radius r_a extends from the electric field axis of symmetry to one of the two radial extrema r_{ex} (of a stable trajectory of interest) and serves to size, or set a scale, of the trajectory.

2) Stable trajectories exist over the wide ranges of Ω and $\mathcal{E}_a$ investigated for all three radial distributions of electric field when Ω and $\mathcal{E}_a$ are of like algebraic sign. These trajectories can be classified into four regions of R_{ex}-, $\mathcal{E}_a$-space according to their resemblance to various cycloidal shapes. Here $R_{ex} \equiv r_{ex}/r_a$. The four regions are separated by two intersecting curves: the curve, or locus of points, where $\mathcal{E}_a = 0$, and the curve through the locus of $dR_{ex}/d\Omega = 0$ points.

3) The motion can be broken into an azimuthal drift about the field axis of symmetry and a net cyclotron motion. The spacial pattern of the cyclotron component is close to that of a circle over nearly all of the ranges of Ω and $\mathcal{E}_a$ of this study. The assumption of a constant cyclotron motion over this pattern is most accurate for the $E \propto r$-field distribution, almost as good for the constant E case, and still less accurate but very reasonable for the $E \propto r^{-1}$ case.

4) The ratio of cyclotron frequency to qB/m equals one where Ω equals zero. This ratio deviates more and more from unity as $|\Omega|$ is increased. It is independent of $\mathcal{E}_a$ however for the case of $E \propto r$. For ions it usually increases with increase of $\mathcal{E}_a$ for the cases of constant E and $E \propto r^{-1}$.

5) The ratio D of guiding center velocity to $-E_a/B$ is dependent on both $\mathcal{E}_a$ and Ω for all three E-field distributions. In the range of most interest in the plasma-heating devices of [5]-[9], the cyclotron frequency ratio varies within the interval of 1.0 to 1.6 and D varies within 0.5 to 1.2. As Ω approaches 0, D approaches 1 and the trajectories reduce to those in a rectilinear $\vec{E} \times \vec{B}$ field.

6) Inclusion of a centrifugal force term in a guiding-center approximation of drift velocity usually makes some improvement over consideration of E/B alone. The deficiency of this approximation is most apparent where the span of the cyclotron motion is large compared to the radius of curvature of the true drift motion.

References

[1] A. W. Hull, "The effect of a uniform magnetic field on the motions of electrons between coaxial cylinders," *Phys. Rev.*, vol. 18, pp. 31-57, 1921.

[2] A. von Engel, *Ionized Gases*. Oxford, England: Clarendon Press, 1965.

[3] J. R. Roth, "Ion heating mechanism in a modified penning discharge," NASA TN D-6985, 1972.

[4] G. X. Kambic, "Determination of the radial potential profile in the modified penning discharge with a heavy ion beam probe," *IEEE Trans. Plasma Sci.*, vol. PS-4, pp. 1-5, 1976.

[5] K. Boyer, J. E. Hammel, C. L. Longmire, D. Nagle, F. L. Ribe, and W. B. Riesenfeld, "Theoretical and experimental discussion of ixion, a possible thermonuclear device," in *Proc. 2nd United National Conf. on Peaceful Uses of Atomic Energy*, vol. 31. Geneva, Switzerland: United Nations, 1958, pp. 319-324.

[6] J. M. Wilcox, "Review of high-temperature rotating-plasma experiments," *Rev. Mod. Phys.*, vol. 31, no. 4, pp. 1045-1051, Oct. 1959.

[7] I. Alexeff, W. D. Jones, R. V. Neidigh, W. F. Peed, and W. L. Stirling, "Plasma heating and burnout in beam-plasma interaction," in *Plasma Physics and Controlled Nuclear Fusion Research* Vienna, Austria: Int. Atomic Energy Agency, vol. II, 1966, pp. 781-799.

[8] D. R. Sigman and J. J. Reinmann, "Steady-state hot-ion plasma produced by crossed electric and magnetic fields," NASA TM X-2783, 1973.

[9] J. J. Reinmann, M. R. Lauver, R. W. Patch, S. J. Posta, A. Synder, and G. Englert, "Hot ion plasma heating experiments in SUMMA–Super-conducting magnetic mirror apparatus," *IEEE Trans. Plasma Sci.*, vol. PS-3, pp. 6-14, 1975.

[10] A. I. Morozov and L. S. Solov'ev, "Motion of charged particles in electromagnetic fields," in *Reviews of Plasma Physics*, vol. II. M. A. Leontovich, Ed, New York: Consultants Bureau, 1966, pp. 201-297.

[11] H. Grad, "The guiding center plasma," in *Proceedings of Symposia in Applied Mathematics*, vol. XVII. Providence, RI: Amer. Math. Soc., 1967, pp. 162-248.

[12] G. W. Englert, J. J. Reinmann, and M. R. Lauver, "Interpretation of neutral particle analyzer measurements on plasma having azimuthal drift," *Plasma Phys.*, vol. 17, pp. 609-621, July-Aug. 1975.

[13] G. W. Englert, R. W. Patch, and J. J. Reinmann, "Model for interpreting doppler broadened optical line emission measurements on axially symmetric plasmas," *Plasma Phys.*, vol. 20, pp. 451-473, May 1978.

[14] R. W. Patch and M. R. Lauver, "Ion temperatures in HIP-1 and SUMMA from charge-exchange neutral optical emission spectra," NASA TM X-73471, 1976.

[15] W. K. H. Panofskv and M. Phillips, *Classical Electricity and Magnetism*, 2nd ed. Reading, MA: Addison-Wesley, 1962, p. 65.

[16] L. Spitzer, *Physics of Fully Ionized Gases*. New York: Interscience, 1956, pp. 2-4.

[17] M. A. Jenkins and J. F. Traub, "A three-stage algorithm for real polynomials using quadratic iteration," *SIAM J. Numer. Anal.*, vol. 7, no. 4, pp. 545-566, Dec. 1970.

[18] F. B. Hildebrand, *Introduction to Numerical Analysis*. New York: McGraw-Hill, 1956, p. 447.

[19] B. O. Peirce, *A Short Table of Integrals*. Boston, MA: Ginn, 1929, p. 23.

[20] V. I. Krylov, *Approximate Calculation of Integrals*, A. H. Stroud, transl. New York: Macmillan, 1962, p. 120.

[21] H. Eves, "Curves and surfaces," *Handbook of Tables for Mathematics*. R. C. Weast and S. W. Salby, Eds. Cleveland, OH: Chemical Rubber Co., 1970, p. 520.